

NetworkX를 사용한 토러스의 그래프 이론적 분석

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Graph Theoretical Analysis of Torus with NetworkX

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Abstract

The multidimensional analysis of torus graphs, constructed through distinct methodologies, presents a rich context to explore and comprehend their intrinsic graph-theoretical properties. This paper embarks on a journey through the mathematical and topological landscape of torus graphs, utilizing two construction paradigms: the structured, grid-like Standard Torus and the stochastically formed Randomized Torus. Leveraging the computational prowess of NetworkX, a Python library, this study investigates key graph properties such as connectivity, Eulerian nature, and node degree distribution within the constructed torus graphs. A comparative analysis between the structured and randomized models illuminates the implications of the construction methodology on the resultant graph-theoretical properties, offering insights into their applicability and performance across diverse network scenarios. Through the lens of these models, the study further explores potential applications and limitations in various domains, including communication networks, social networks, and more. This exploration not only enhances our understanding of torus graphs but also underscores the versatility of graph theory in modeling and analyzing complex networked systems across multiple disciplines. This concise analysis, while providing a foundational framework, also paves the way for future research exploring other topological spaces and graph models across varied construction paradigms.

1. Introduction

A torus, defined mathematically as a surface of revolution generated by revolving a circle in three-dimensional space about an axis coplanar with the circle, exhibits fascinating properties when observed through the lens of graph theory. Graph theory provides a robust framework to explore the structural and topological properties of various shapes and networks, including the torus. In this context, a graph can be understood as a collection of vertices (or nodes) and edges (or arcs) that connect pairs of vertices. The torus can be envisaged as a graph by considering points (vertices) on its surface and connecting them with edges, embodying the relationships or paths between these points.

NetworkX, a Python library, enables the creation, manipulation, and study of the structure and dynamics of complex networks of interaction [1]. It provides tools to explore and visualize graph properties, thereby serving as a powerful computational resource for graph theorists and network scientists. Particularly, it offers functionalities to investigate various graph properties, such as connectivity, paths, and others, which are crucial in the comprehensive analysis of a torus graph.

In the subsequent sections, we delve into the construction and examination of torus graphs, employing two distinct approaches: a structured, grid-like formation and a randomized node generation. The former represents a conventional, systematic representation, while the latter offers a stochastic model, thereby providing a nuanced perspective on the implications of node placement and edge formation within the torus graph. This paper aims to illuminate the graph theoretical properties of a torus, exploiting the computational capabilities of NetworkX, and subsequently presents a comparative analysis of the two aforementioned models.

2. Construction of a Torus Graph

The construction of a torus graph can be approached via two distinct methodologies: a Standard Torus Graph and a Randomized Torus Graph, each presenting unique attributes and challenges in exploring graph theoretical properties.

2.1. Standard Torus

The Standard Torus Graph is typically represented in a structured, grid-like format. It is derived from a 2D grid graph, which essentially consists of nodes arranged in a rectangular lattice, wherein each node is connected to its adjacent nodes. The

transformation from a 2D grid to a 3D torus involves a mathematical mapping, utilizing trigonometric functions to project nodes into a toroidal shape, adhering to the equation:

$$(x, y, z) = ((R + r \cos \theta) \cos \phi, (R + r \cos \theta) \sin \phi, r \sin \theta)$$

where θ and ϕ are angles which parametrically represent the torus in 3D space, and R and r are the major and minor radii respectively [2]. Nodes are strategically connected to maintain grid consistency, ensuring each node is connected to its adjacent nodes, even at the boundaries, thereby preserving the topological integrity of the torus. The visual representation elucidates the structured, symmetrical nature of the constructed graph Figure 1.

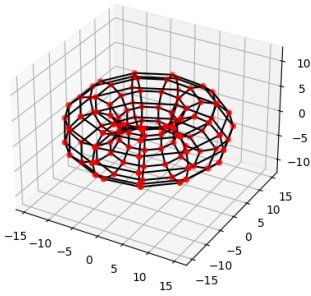


Figure 1. Constructed Graph of Torus

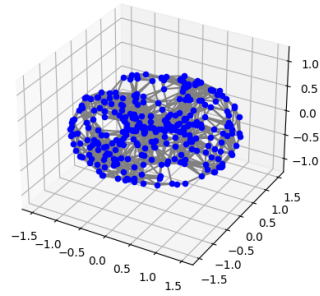


Figure 2. Randomized Graph of Torus

2.2. Randomized Torus

Conversely, the Randomized Torus Graph is constructed by generating nodes at random coordinates within a predefined 3D space, forming a seemingly arbitrary toroidal configuration. Edges are formed based on a proximity criterion: nodes within a certain distance threshold are connected by an edge. Mathematically, the distance between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) in 3D space is given by:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

If d is less than a predefined threshold, an edge is formed between the nodes. This approach, while less structured than the grid-based model, provides insights into the behavior and properties of more chaotic, less predictable networks, thereby serving as a contrasting model to the standard torus. The visual representation of this model would exhibit a less symmetrical and more sporadic node and edge distribution Figure 2.

Both models, despite their contrasting construction methodologies, provide a fertile ground for exploring and contrasting the implications of structure and randomness on the graph-theoretical properties of toroidal graphs, which will be further explored in the subsequent sections.

3. Graph Theoretical Properties

The topological analysis of the torus graphs, both standard and randomized, affords a deep dive into understanding their inherent graph-theoretical properties, particularly focusing on connectivity, Eulerian nature, and node degree distribution.

3.1. Connectivity Analysis

Connectivity in a graph refers to the graph's robustness as a network, measured by its ability to remain connected even when some vertices or edges are removed [3]. A graph is termed connected if there exists a path between every pair of vertices. Both the standard and randomized torus graphs, despite their contrasting constructions, exhibit intriguing connectivity characteristics that might be influenced by their respective edge formation methodologies.

3.2. Eulerian Graph Analysis

An Eulerian graph is defined as a connected graph where every vertex has an even degree, thereby enabling the formation of an Eulerian circuit, a cycle traversing each edge exactly once and returning to the starting vertex [4]. The structured nature of the standard torus graph inherently facilitates an Eulerian characteristic, as demonstrated in the implementation. However, the randomized torus graph introduces a degree of unpredictability to this property, offering a rich context for exploring the implications of randomization on Eulerian nature.

3.3. Degree of Nodes

The degree of a node in a graph is defined as the number of edges incident to it. Investigating the degree distribution of nodes within the torus graphs can unearth insights into their structural and topological nuances. For instance, the standard torus, owing to its structured construction, inherently possesses a uniform degree distribution, with each node maintaining an even degree. In contrast, the randomized torus, due to its stochastic node placement and edge formation, might exhibit a varied degree distribution, potentially introducing nodes with odd degrees, as elucidated in the provided implementation.

Through these lenses, this paper will explore the intrinsic graph-theoretical properties of torus graphs, further elucidating the implications of structural versus random construction on these properties. The subsequent section will undertake a comparative analysis of these properties between the two construction methodologies, aiming to discern their respective impacts and potential applications in various domains.

4. Comparative Analysis

The juxtaposition of the Standard and Randomized Torus graph models highlights distinct variances in their graph-theoretical properties, eliciting a deeper comprehension of the implications of different construction methodologies on the topological characteristics of the resultant graphs.

4.1. Structural Versus Stochastic Nodes and Edges

The Standard Torus, with its methodical and symmetrical structure, provides a model where nodes and edges are uniformly distributed, ensuring stable connectivity, consistent node degree, and a guaranteed Eulerian nature. The uniformity and predictability of this model might find applications in network

design where reliability and regularity are paramount, such as in communication networks or grid computing [5].

On the contrary, the Randomized Torus, with its stochastic node placement and proximity-based edge formation, introduces an element of irregularity and unpredictability in connectivity and node degree distribution. This model may be indicative of real-world networks where nodes (e.g., users, devices, etc.) are not uniformly distributed or where connections are formed based on proximity or other localized criteria, such as social networks or ad-hoc communication networks [6].

4.2. Connectivity and Robustness

While both models exhibit overall connectivity, the structured approach of the Standard Torus ensures a uniform and thus, potentially more robust network under conditions of node or edge failures. The Randomized Torus, however, may exhibit vulnerabilities or variations in network robustness due to its non-uniform node and edge distribution, impacting its performance under similar conditions of stress or failure.

4.3. Eulerian Nature and Path Traversal

The Eulerian characteristics of the two models diverge due to the inherent even degree of all nodes in the Standard Torus, as opposed to the varied node degree in the Randomized Torus. The implications extend to path traversal algorithms and network routing protocols, where the Standard Torus ensures the existence of Eulerian circuits, thereby optimizing certain path traversal and data transmission methodologies.

4.4. Applications and Limitations

The deterministic nature of the Standard Torus makes it suitable for applications demanding reliable, consistent network topologies, while the Randomized Torus may find relevance in modeling and analyzing networks where nodes are irregularly distributed or where connections are unpredictably formed.

In summary, the exploration and comparative analysis of the two torus models illuminate the significance of construction methodologies on the resultant graph-theoretical properties, providing a comprehensive framework to understand, model, and analyze various network topologies and their potential applications across diverse domains.

5. Conclusion

The exploration of the torus through structured and randomized graph models elucidates the intricate relationship between construction methodologies and resultant graph-theoretical properties. The Standard Torus, with its structured grid, offers uniformity and predictability, thereby assuring stable and consistent network properties. This model can serve as a pivotal framework in domains requiring steadfast and reliable network structures, such as computational grids or structured peer-to-peer networks [5].

Conversely, the Randomized Torus, through its stochastic node placement and edge formation, provides insights into networks

characterized by unpredictability and irregularity. This model may be particularly pertinent in understanding and analyzing real-world networks, where the distribution of nodes and formation of connections are influenced by various stochastic factors, such as in social or biological networks [6].

In practical applications, the choice between structured and randomized network models often hinges on the specific requirements and constraints of the domain. Structured networks may be preferred in scenarios demanding reliability and optimal path traversal, while randomized models might offer a more accurate representation of networks in natural or social systems, where nodes and connections manifest with inherent unpredictability.

While this study provides a foundational framework for understanding and analyzing torus graphs via distinct methodologies, future research might delve deeper into exploring other topological spaces and graph models, investigating their properties under varied construction paradigms, and discerning their applicability and limitations across diverse fields.

In closing, the analytical exploration of torus graphs, facilitated by the computational capabilities of NetworkX, underscores the versatility and profundity of graph theory in modeling, analyzing, and understanding complex systems across varied domains, thereby offering a robust toolset for researchers and practitioners in network science, computer science, and beyond.

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